PLDSC: univariate and bivariate LD Score regression

Methods and implementation notes, 29 October 2026.
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PLDSC estimates SNP heritability from one GWAS and genetic correlation from two GWAS using LD Score regression (LDSC). It also contains research tools for assessing LD-score mismatch in multi-ancestry meta-analyses. This report explains the ordinary LDSC fits, their inputs and computational design. The research results remain in `docs/MULTI_ANCESTRY.md`; the runnable examples and full API guide are in `docs/LDSC.md`.

1 Inputs and targets

For variant j in trait t, let $z_{tj} = b_{tj}/s_{tj}$ be the signed GWAS effect divided by its standard error, and let N_{tj} be the appropriate effective sample size. For a reference LD correlation r_{jk}, define the LD score over a reference variant set $\mathcal{M}$ of size M as

$$\ell_j = \sum_{k \in \mathcal{M}} r_{jk}^2. \tag{1}$$

The reference must match the populations and genome build of the GWAS well enough for the intended analysis, and the score for each regression variant must be associated with the correct reference row. M counts the variants summed into the score; it is not, in general, the number of GWAS rows left after QC. PLDSC fits one common LD-score vector for the two traits, so a single reference must be appropriate for both.

The targets are the SNP heritabilities h_1^2, h_2^2 and the additive genetic covariance g_{12} for the modeled variant set and scale. Their genetic correlation is

$$r_g = \frac{g_{12}}{\sqrt{h_1^2 h_2^2}}. \tag{2}$$

This is a correlation of genetic components, not a phenotypic correlation or a causal effect. A binary trait is not automatically transformed to the liability scale. Reference mismatch, effect-size architecture, selection and imperfect LD estimates can alter the interpretation of the fitted slope.

2 The regressions

Under the standard LDSC approximation, with standardized effects having roughly constant variance and covariance across variants, the conditional moments follow Equations (3) and (4) [1, 2]:

$$\mathbb{E}[z_{tj}^2 \mid \ell_j] \approx a_t + \frac{N_{tj}\ell_j}{M} h_t^2, \tag{3}$$

$$\mathbb{E}[z_{1j} z_{2j} \mid \ell_j] \approx a_{12} + \frac{\sqrt{N_{1j}N_{2j}}\,\ell_j}{M} g_{12}. \tag{4}$$

The slope of Equation (3) estimates h_t^2; the slope of Equation (4) estimates g_{12}. PLDSC fits both marginal regressions and the cross-trait regression on the *same common rows* for a bivariate analysis, and then uses Equation (2). A separate one-trait fit can have a different h^2 because it can retain different variants. The two GWAS effects must count the same allele before their signed z-scores are multiplied.

The intercepts are free by default. The marginal intercept can reflect confounding, genomic-control effects or model/reference mismatch; the cross-trait intercept can also reflect sample overlap and correlated confounding. These sources are not identified separately by the intercept. Fixing the intercept is an assumption, not a diagnostic: in the bivariate fit `constrain_intercept` fixes only a_{12}, leaving a_1, a_2 free. A misspecified fixed intercept can bias the slopes.

2.1 Weights and uncertainty

PLDSC uses one-step iteratively reweighted least squares. Let $x_{tj} = N_{tj}\ell_j/M$ and $L_j = \max(\ell_j, 1)$. A marginal fit starts with $\mu_{tj} = 1$ and, after each fit, replaces it by $\max(a_t + h_t^2 x_{tj}, 1)$. Its weights are

$$w_{tj} = \frac{1}{2\mu_{tj}^2 L_j}. \tag{5}$$

For the cross-trait fit, the fitted marginal means give the approximate bivariate-normal variance of $z_{1j}z_{2j}$. With fitted cross mean q_j, PLDSC uses

$$w_{12,j} = \frac{1}{L_j \max(\mu_{1j}\mu_{2j} + q_j^2, 10^{-6})}. \tag{6}$$

The first cross fit uses $q_j = 0$ when its intercept is free, or the fixed intercept otherwise; subsequent fits update q_j. By default there are two weight updates after the initial fit. This is a one-step estimator: PLDSC does not reproduce every option, including the original software's two-step intercept estimator, of the standalone LDSC implementation.

Standard errors use a delete-a-block jackknife over contiguous variants in genomic order. If $\widehat{\theta}_{(-b)}$ is the estimate after deleting block b, with mean $\overline{\theta}_{(-\cdot)}$ across B deletions,

$$\widehat{\mathrm{se}}(\widehat{\theta}) = \left[\frac{B-1}{B}\sum_{b=1}^{B}(\widehat{\theta}_{(-b)} - \overline{\theta}_{(-\cdot)})^2\right]^{1/2}. \tag{7}$$

Each deletion refits the weights; a bivariate deletion refits both marginals and the cross-trait regression before calculating r_g. The defaults are 100 blocks for one trait and 200 for two, capped by the number of regression variants. For r_g, the point estimate is undefined if either marginal estimate is non-positive; its standard error is undefined if a required replicate is invalid or fewer than two blocks are available. Estimates are not clipped to their theoretical ranges.

3 From files to estimates

Figure 1 summarizes the file workflow. Allelign reads each GWAS, checks its columns and sample sizes, filters rows, and harmonizes alleles to a sorted `.bim` or `.pvar` reference. Defaults remove non-finite effects, non-positive standard errors or sample sizes, duplicated identifiers, and variants with N below 70% of the median. Where available, MAF below 0.01 and INFO below 0.7 are filtered. Ambiguous A/T and C/G sites are dropped by default, and effect signs are flipped where alignment requires it. QC and alignment counts are returned for each trait.

The wrappers then apply the per-trait cap $z_{tj}^2 \le \max(80, 0.001N_{tj})$; a bivariate row must pass both caps. They do not automatically remove long-range LD regions or check genome builds. Such selection, liftover and reference suitability are the analyst's responsibility. A numerical `n_eff` passed as a read option replaces every per-variant sample size. For a case-control study, a common choice is

$$N_{\mathrm{eff}} = \frac{4}{1/N_{\mathrm{case}} + 1/N_{\mathrm{control}}}. \tag{8}$$

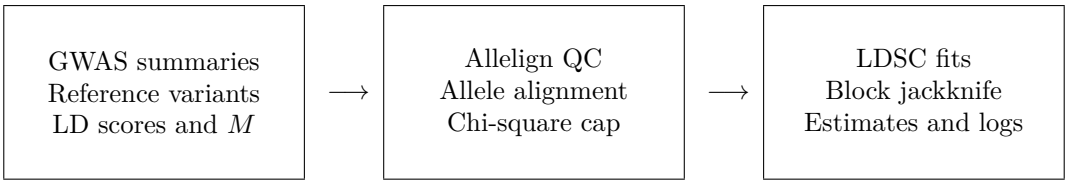

Figure 1: The file-based workflow for one or two traits. Bivariate fits use common, aligned variants passing the cap for both traits.

For compatible meta-analyses of non-overlapping cohorts, the sum of cohort-specific effective sizes can differ from applying Equation (8) to pooled counts; choose the scale appropriate to the GWAS.

Table 1 separates the file-based and preprocessed-array paths. The latter do not perform QC, harmonization, filtering or sorting. The `ldsc_rg` array function takes LDpred3-standardized effects; raw per-allele GWAS effects belong in the file wrapper or must first be converted to signed z-scores for `ldsc_rg_z`. For example, with file paths `gwas1`, `gwas2`, and `ref`, and reference-aligned `scores`, the file-based calls are:

```
h2 = ldsc_h2_sumstats(gwas1, ref, scores, m_snps=M)
rg = ldsc_rg_sumstats(gwas1, gwas2, ref, scores,
                      m_snps=M, ncores=4)
print(h2.estimate.h2, rg.estimate.rg, rg.estimate.rg_se)
```

Both functions are imported from `pldsc`. Their results also contain QC and alignment logs; the complete, runnable example is in the guide.

Table 1: Main entry points and their inputs. The file wrappers return QC and alignment logs alongside the estimate.

Function	Input and result
`ldsc_h2_sumstats`	One GWAS and reference: SNP heritability and SE.
`ldsc_rg_sumstats`	Two GWAS and reference: genetic covariance, correlation and correlation SE.
`ldsc_h2`	Aligned arrays of z^2, LD scores and sample sizes.
`ldsc_rg_z`	Aligned arrays of signed z-scores, LD scores and sample sizes.

4 Scores, computation and evidence

`ld_scores` accepts a dense correlation matrix or disjoint indexed blocks, including int8 and low-rank representations. Low-rank contractions avoid materializing the full dense matrix. Block scores omit between-block LD, and the optional finite-reference correction can yield non-positive scores. The regressions require positive scores; silently clipping them changes the analysis. Score construction and the choice of M must match the intended reference, window and variant set.

The regression kernels were adapted from LDpred3 and bipred and are designed for very fast repeated fits. With optional Numba compilation, delete-block fits traverse retained ranges without making a full copy for each deletion. Independent bivariate deletions can run in parallel; poorly conditioned fits fall back to the reference numerical path. Without Numba, a NumPy implementation remains available. Compilation affects the first call, while elapsed time and memory depend on the workload and hardware. The repository includes a paired warm timing driver; no quantitative speedup is asserted here without a corresponding measured run.

The tests include an exact univariate regression line, dense and low-rank score checks, near-singular fits, allele flips and file-workflow checks, and comparisons with LDpred3 and bipred when those siblings are installed. These are implementation and consistency checks, not evidence of real-data calibration. The synthetic example exercises the API but does not model LD among GWAS statistics.

5 Relation to the population-structure research

The ordinary regressions above use one LD-score vector. The multi-ancestry research instead derives a meta-analysis score containing within- and cross-ancestry LD and examines the consequences of using a single-ancestry reference. Its ancestry-pair effect correlation is not the cross-trait r_g in Equation (2). In the current panel experiments, reported κ values are panel plug-in slope ratios under stated assumptions; finite-reference noise can affect both scores, so these are not calibrated estimates of absolute heritability bias. The definitions, assumptions, experiments and results are recorded in `docs/MULTI_ANCESTRY.md`.

References

[1] Bulik-Sullivan BK, Loh PR, Finucane HK, et al. LD Score regression distinguishes confounding from polygenicity in genome-wide association studies. *Nature Genetics* 47, 291–295 (2015). doi:10.1038/ng.3211.

[2] Bulik-Sullivan B, Finucane HK, Anttila V, et al. An atlas of genetic correlations across human diseases and traits. *Nature Genetics* 47, 1236–1241 (2015). doi:10.1038/ng.3406.